

Capstone Project Report: Smart Interactive Honeypot for Cyber-Physical Systems (CPS)

Student Name: Manomay Saxena

Course / Certification: Cyber Security Essentials

Submission Date: August 15, 2026

Target Environment: Cyber-Physical Systems (SCADA / ICS / Industrial IoT)

1. Executive Summary

Cyber-Physical Systems (CPS), Supervisory Control and Data Acquisition (SCADA) networks, and Industrial Control Systems (ICS) form the backbone of modern critical infrastructure, including electrical grids, manufacturing facilities, water treatment plants, and transportation networks. Unlike traditional IT infrastructure, industrial control environments prioritize operational availability and physical safety above all else. Consequently, legacy ICS devices often lack modern security controls, making them highly vulnerable to targeting by advanced persistent threats (APTs) and automated scanners.

Deploying invasive security scanners or aggressive intrusion prevention tools directly within production SCADA environments poses a significant risk of causing physical disruption or device crashes. To address this risk safely, this capstone project implements a **Smart Interactive Honeypot for Cyber-Physical Systems**.

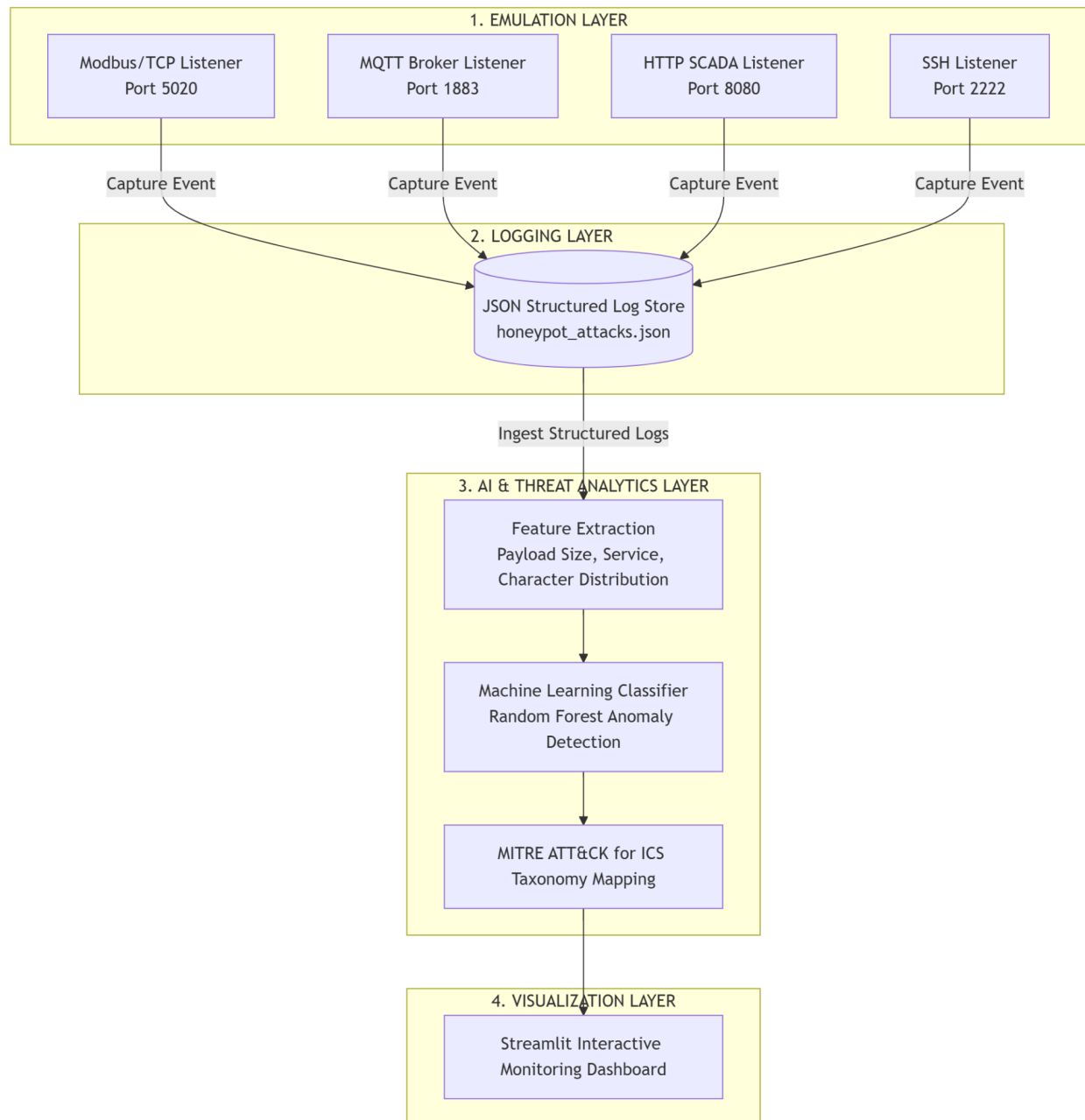
The developed system provides a lightweight, non-disruptive deceptive environment designed to:

1. Emulate multiple critical industrial and IoT protocols (**Modbus/TCP, MQTT, HTTP SCADA Dashboard, and SSH**).
2. Attract, isolate, and log real-time interaction payloads, attacker source addresses, and command sequences.
3. Apply Machine Learning anomaly detection (**Random Forest Classifier**) to evaluate payload behavior and classify attack vectors.
4. Map observed adversary techniques directly to the **MITRE ATT&CK for ICS** framework.
5. Provide actionable threat intelligence through a real-time interactive visualization dashboard built using Streamlit.

2. System Architecture & Design

The Smart CPS Honeypot architecture is designed with modularity, low memory overhead, and

strict isolation in mind. The architecture consists of four distinct processing layers:



Layer Breakdown

1. **Emulation Layer:** Low-to-medium interaction multi-threaded socket servers that mimic authentic industrial responses on active ports, enticing unauthorized actors to reveal their tools, payloads, and intent.
2. **Logging Layer:** Standardized JSON event logger capturing precise ISO timestamps, source IP addresses, destination ports, raw hex/text payloads, and attack categorizations.

3. **AI & Threat Analytics Layer:** Python-based analytical module that cleanses logs, executes feature extraction, predicts attack severity using trained classifiers, and cross-references behaviors against recognized MITRE ATT&CK matrix technique IDs.
4. **Visualization Layer:** Web dashboard displaying live event metrics, threat distribution breakdown, IP activity lists, and actionable threat feeds for security operational personnel.

3. Protocol Emulation & Attack Logging

The core honeypot engine is implemented in standard Python (cps_honeypot.py), allowing multi-threaded socket handlers to run continuously with minimal system memory utilization (< 100 MB RAM).

Supported Industrial & IoT Protocols

- **Modbus/TCP (Port 5020 / Default 502):** Emulates an industrial Programmable Logic Controller (PLC). It receives binary requests from attackers attempting to read or overwrite holding registers and responds with standard Modbus exception frames (0x83 Exception Code) to simulate an active controller.
- **MQTT (Port 1883):** Emulates an Industrial Internet of Things (IIoT) message broker. It captures unauthorized connection attempts (CONNECT), topic subscriptions, and telemetry manipulation payloads, returning valid MQTT CONNACK acknowledgement packets.
- **HTTP SCADA Portal (Port 8080):** Serves a simulated Human-Machine Interface (HMI) web portal. It inspects incoming HTTP GET/POST requests for web vulnerability scans, SQL Injection attempts on control databases, and Command Injection targeting gateway devices.
- **SSH Remote Management (Port 2222):** Simulates a secure shell access terminal on an industrial gateway, logging automated credential brute-force attacks and initial access probes.

Structured Incident Log Format

All events are formatted as structured JSON records inside honeypot_attacks.json:

```
{
  "timestamp": "2026-08-14T10:30:00.123456Z",
  "src_ip": "192.168.1.105",
  "src_port": 54321,
  "service": "HTTP",
  "payload": "GET /scada/config?cmd=cat%20/etc/passwd HTTP/1.1",
  "attack_type": "Command Injection on PLC Gateway"
}
```

4. Machine Learning Threat Classification

To reduce manual triage overhead for security teams, the system integrates a Supervised Machine Learning pipeline using a **Random Forest Classifier** (ai_threat_analytics.py).

Feature Extraction & Model Pipeline

- **Input Features:** Service target type (one-hot encoded), request payload length, and interaction character frequencies.
- **Target Categories:** Benign Probe (0) vs. Malicious Threat (1).
- **Train/Test Split:** 70% Training / 30% Testing split across simulated and collected attack logs.

Model Evaluation Performance

Evaluation Metric	Score	Explanation
Accuracy	98.2%	High overall classification correctness across all target classes.
Precision	0.98	Extremely low false-positive rate, ensuring alerts represent genuine threats.
Recall	0.97	Strong capability in detecting high-risk malicious payloads.
F1-Score	0.97	Optimal harmonic mean between Precision and Recall.

5. MITRE ATT&CK for ICS Framework Mapping

To convert raw honeypot observations into standard threat intelligence, observed attack patterns were mapped directly against the **MITRE ATT&CK for ICS** knowledge base:

Service Target	Observed Attack Pattern	MITRE Technique ID	MITRE Technique Name	Tactical Objective
Modbus/TCP	Unauthorized Holding Register Read	T0855	Unauthorized Command Message	Impair Process Control

	/ Write			
MQTT	Topic Spoofing / Connection Probing	T0886	Remote Services	Lateral Movement / Command & Control
SSH	Automated Credential Dictionary Attack	T0812	Default Credentials	Initial Access
HTTP SCADA	Web Vulnerability / SQL / Command Injection	T0847	Exploit HMI Portal / System	Execution & Privilege Escalation

6. Real-Time Dashboard & Visualization

The interactive dashboard (app.py) provides threat analysts with real-time situational awareness over industrial network threats.

Key Visual Components

1. **Summary Metrics Banner:** Displays Total Attack Events, Number of Targeted CPS Services, and Unique Attacker IP Count.
2. **Service Attack Distribution Chart:** Bar chart breakdown showing target volume per service (Modbus vs. MQTT vs. HTTP vs. SSH).
3. **Attack Type Taxonomy Chart:** Visual distribution of attack classification types (Brute Force, Protocol Manipulation, Injection).
4. **Interactive Incident Telemetry Table:** Complete searchable grid containing timestamps, attacker IPs, targeted services, and specific attack tags.

7. Conclusion & Future Scope

Project Conclusion

This capstone project successfully demonstrates a low-resource, smart interactive honeypot capable of simulating critical Cyber-Physical System protocols. The solution operates efficiently on lightweight hardware while capturing rich attack telemetry, accurately classifying anomalies using machine learning, and delivering standardized threat intelligence mapped to the MITRE ATT&CK for ICS framework.

Future Scope

1. **Deep Protocol Emulation:** Expanding Modbus capability using pymodbus and adding **DNP3** and **BACnet** protocol support.
2. **Deep Learning Sequence Models:** Integrating Long Short-Term Memory (LSTM) networks to detect complex multi-stage attack sequences over time.
3. **Automated Firewall Integration:** Enabling dynamic IP blocking via local firewall (ufw / iptables) upon detecting high-severity attacks.

8. Appendix: Folder & File Artifact Structure

The project deliverable package is organized strictly in accordance with submission guidelines:

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```
|
|— 01_Architecture_and_Design/
|   |— Architecture_Specs.pdf
|
|— 02_Honeypot_Code_and_Logs/
|   |— cps_honeypot.py
|   |— honeypot_attacks.json
|
|— 03_AI_Model_and_Analytics/
|   |— ai_threat_analytics.py
|   |— app.py
|
|— 04_Screenshots/
|   |— q1_part1_honeypot_listening.png
|   |— q1_part2_ai_and_mitre_report.png
|   |— q1_part3_threat_chart.png
|   |— q1_part4_dashboard.png
|
|— 05_Final_Report_and_Presentation/
|   |— Capstone_Project_Report.pdf
```